20/7/2026

Assessment-1

85/100

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with K-Nearest Neighbours algorithm is applied for binary classification. The dataset consists of 1800 samples per class, with 20 % of the data reserved for testing. During evaluation, the model reports 210 True positives (TP), 190 True negatives (TN), 30 false positives (FP). Based on the given information, determine the missing false Negatives (FN) value, and then calculate Accuracy, Precision, sensitivity, and Specificity of the classifier, explaining its effectiveness in medical diagnosis.

Sol → Given,

Total samples per class = 1800

Binary classes = Malignant and Benign

Total dataset = 1800 + 1800 = 3600

Test data = 20 % of 3600

= 720 Samples

Confusion matrix values:

→ True Positives (TP) = 210

→ True Negatives (TN) = 190

→ false Positives (FP) = 30

→ false Negatives (FN) = ?

Since the test set contains 720 samples

$$TP + TN + FP + FN = 720$$

$$210 + 190 + 30 + FN = 720$$

$$430 + FN = 720$$

$$FN = 720 - 430$$

$$\boxed{FN = 290}$$

(i) Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{210 + 190}{720}$$

$$= \frac{400}{720} = 0.556$$

$$\boxed{\text{Accuracy} = 55.56\ \%}$$

(ii) Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{210}{210 + 30}$$

$$= \frac{210}{240} = 0.875$$

$$\boxed{\text{Precision} = 87.5\ \%}$$

(iii) Sensitivity

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$= \frac{210}{210 + 290}$$

$$= \frac{210}{500}$$

$$= 0.42$$

$$\boxed{\text{Sensitivity} = 42\ \%}$$

(iv) Specificity:

$$\text{Specificity} = \frac{TN}{TN+FP}$$

$$= \frac{190}{190+30} = \frac{190}{220}$$

$$= 0.8636$$

$$\boxed{\text{Specificity} = 86.36\%}$$

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4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs:

(a) Epoch 1 : Training Loss = 2.8, Validation loss = 2.4
(b) Epoch 2 : Training loss = 2.5, validation loss = 22
(c) Epoch 3 : Training loss = 2.2, validation loss = 2.0
(d) Epoch 4 : Training loss = 2.0, validation loss = 2.1
(e) Epoch 5 : Training loss = 1.8, validation loss = 2.3
(f) Epoch 6 : Training loss = 1.6, validation loss = 2.6
(g) Epoch 7 : Training loss = 1.5, validation loss = 2.9
(h) Epoch 8 : Training loss = 1.3, validation loss = 3.2

(i) Identify the epoch at which the model begins to overfit.

(ii) Suggest two effective techniques to reduce overfitting in this model.

(iii) Discuss how overfitting affects model's generalisation ability on unseen data.

Sol → Given data,

Epoch	Training loss	Validation loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

(i) Epoch at which model begins to overfit.

The model starts overfitting at Epoch 4.

Reason: Upto Epoch 3, both training loss and validation loss decrease. From Epoch 4 onwards, the training loss continues to decrease, but the validation loss starts increasing. This indicates the model is learning the training data too well and performing worse on unseen data.

(ii) Two Techniques to reduce overfitting.

1. Early stopping: Stop training when the validation loss starts increasing.

2. Dropout / Regularization: Randomly deactivate neurons or add a penalty to reduce model complexity and prevent memorization.

(iii) Effect of overfitting on generalization.

Overfitting causes the model to memorize the training data instead of learning general patterns. As a result, it performs well on the training dataset but gives poor accuracy and higher errors on new, unseen test data. Therefore, the model's generalization ability decrease, making it unreliable for real-world predictions.